

Chapter 2: Statistical Distributions for Financial Markets

Complete Pedagogical Architecture

Statistics of Financial Markets

Daniel Traian PELE

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Part I

CHAPTER ARCHITECTURE

1 Hierarchical Chapter Outline

2.1 Introduction — Why Distributions Matter in Finance

- [Y3] Risk measurement motivation, tail risk intuition, model risk
- [MSc] Distribution choice and derivative pricing
- [PhD] Model uncertainty quantification

2.2 From Prices to Log-Returns

- [Y3] Definition, computation, statistical properties
- [MSc] Continuous-time limit, Itô calculus connection
- [PhD] Semimartingale representation

2.3 The Normal Distribution — The Classical Model

- [Y3] PDF, moments, CLT, quantiles (preparation for VaR in Section 2.10)
- [MSc] Maximum entropy characterization, normality tests
- [PhD] Berry–Esseen bounds, rate of convergence

2.4 Stylized Facts of Financial Returns

- [Y3] Fat tails, skewness, volatility clustering
- [MSc] Aggregational Gaussianity, leverage effect, long memory
- [PhD] Cont (2001) taxonomy, fractal structure, multiscaling

2.5 The Student- t Distribution

- [Y3] PDF, degrees of freedom, excess kurtosis, MLE fitting
- [MSc] Location-scale form, AIC/BIC comparison, GARCH- t
- [PhD] Mixture representation, connection to F-distribution, Bayesian interpretation

2.6 Asymmetric Distributions

- [Y3] Skewness effect on downside risk, basic intuition
- [MSc] Hansen’s skewed- t , GED, MLE estimation
- [PhD] Generalized Hyperbolic family, NIG, Barndorff-Nielsen (1977)

2.7 Stable (α -Stable) Distributions [MANDATORY]

- [Y3] Intuition, stability property, tail index α , why variance may not exist
- [MSc] Characteristic function, GCLT, estimation (McCulloch, MLE), financial interpretation
- [PhD] Regular variation, domain of attraction, Lévy processes, subordinators, tempered stable

2.8 Extreme Value Theory (EVT)

- [Y3] Tail behavior intuition, GPD concept (descriptive), extreme events in finance
- [MSc] Fisher–Tippett–Gnedenko theorem, POT method, threshold selection, EVT-based VaR
- [PhD] Hill estimator, mean excess function, second-order conditions, bias–variance trade-off

2.9 Distribution Comparison and Model Selection

- [Y3] Visual comparison, QQ-plots, basic KS test

- [MSc] AIC/BIC, Anderson–Darling, likelihood ratio tests
- [PhD] Vuong test, cross-validation, scoring rules, Diebold–Mariano

2.10 Risk Management Implications

- [Y3] **Define VaR from scratch:** definition, interpretation, graphical intuition, VaR under normality, numerical example, comparison Normal vs. Student- t
- [MSc] **Define ES from scratch,** coherence axioms, Kupiec test, backtesting, Basel framework
- [PhD] Spectral risk measures, elicibility, regulatory capital optimization

2.11 Modern Extensions

- [Y3] GARCH intuition (conceptual only)
- [MSc] GARCH- t , Gaussian mixtures, regime switching
- [PhD] Stochastic volatility with jumps, Lévy-driven models, realized measures

2 Slide Count Allocation

Section	Y3 Slides	MSc Slides	PhD Slides	Total
2.1 Introduction	3	1	1	5
2.2 Prices to Log>Returns	2	1	1	4
2.3 Normal Distribution	4	2	1	7
2.4 Stylized Facts	4	2	2	8
2.5 Student- t Distribution	4	2	1	7
2.6 Asymmetric Distributions	2	3	2	7
2.7 Stable Distributions	4	4	4	12
2.8 Extreme Value Theory	3	4	3	10
2.9 Distribution Comparison	3	2	2	7
2.10 Risk Management	5	3	2	10
2.11 Modern Extensions	1	3	3	7
Conclusions + Homework	2	1	1	4
Total	37	28	23	88

Part II

LAYERED CONTENT STRUCTURE

3 Section 2.1: Introduction — Why Distributions Matter

3.1 Year 3 Core

- **Motivating question:** Why does it matter which probability distribution we use to model financial returns?
- If we assume the wrong distribution, we underestimate the probability of extreme losses
- **Example:** October 19, 1987 — the S&P 500 fell 20.5% in a single day

- Under the Normal assumption, this would be a $\sim 25\sigma$ event with probability $\approx 10^{-135}$
- Yet it happened — suggesting the Normal distribution is inadequate for tail events
- Banks and regulators need to quantify “how bad can it get?” — this requires accurate distributional models
- **Preview:** In Section 2.10 we will formally define **Value-at-Risk (VaR)**, a key risk measure that depends entirely on the distributional assumption
- The distribution choice problem: Normal vs. Student- t vs. Stable vs. EVT

3.2 Master Extension

- Impact on derivative pricing: Black–Scholes assumes log-normality; deviations produce the volatility smile
- Portfolio optimization under non-Gaussian returns: mean-variance is suboptimal
- Regulatory context: Basel II $\rightarrow$ Basel III/IV framework (details in Section 2.10)

3.3 PhD Extension

- Model uncertainty and ambiguity aversion (Ellsberg paradox in portfolio choice)
- Robust risk measures under distributional uncertainty
- Minimax approaches: worst-case distributional analysis

4 Section 2.2: From Prices to Log-Returns

4.1 Year 3 Core

Definition 1 (Log-Return). The **log-return** (continuously compounded return) at time t is:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln P_t - \ln P_{t-1}$$

where P_t is the asset price at time t .

- **Additivity:** Multi-period returns are sums: $r_{t:t+k} = \sum_{i=1}^k r_{t+i}$
- **Approximation:** For small returns, $r_t \approx (P_t - P_{t-1})/P_{t-1}$
- **Statistical motivation:** Additive structure enables standard probability theory

4.2 Master Extension

- Continuous-time limit: $dS/S = \mu dt + \sigma dW \Rightarrow d \ln S = (\mu - \sigma^2/2) dt + \sigma dW$
- Jensen’s inequality: $\mathbb{E}[\ln(P_T/P_0)] \neq \ln(\mathbb{E}[P_T/P_0])$
- Log-return vs. simple return: implications for portfolio aggregation

4.3 PhD Extension

- Semimartingale decomposition and Itô’s lemma
- Quadratic variation and realized volatility: $[X, X]_t = \lim \sum (X_{t_{i+1}} - X_{t_i})^2$
- Non-continuous price processes: jumps and Lévy–Itô decomposition

5 Section 2.3: The Normal Distribution

5.1 Year 3 Core

Definition 2 (Normal Distribution). $X \sim N(\mu, \sigma^2)$ if:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

Key properties:

- $\mathbb{E}[X] = \mu$, $\text{Var}(X) = \sigma^2$
- Skewness = 0, Kurtosis = 3 (excess kurtosis = 0)
- 68–95–99.7 rule
- **Quantiles:** The α -quantile of $N(\mu, \sigma^2)$ is $q_\alpha = \mu + \sigma \Phi^{-1}(\alpha)$; this will be used in Section 2.10 to define Value-at-Risk
- **CLT:** $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$
- **Stability:** $X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$ for independent normals
- **Limitation:** Tails decay as $e^{-x^2/2}$ — too thin for financial data

5.2 Master Extension

- Maximum entropy characterization: among all distributions with given (μ, σ^2) , the normal maximizes Shannon entropy
- Normality tests: Jarque–Bera, Shapiro–Wilk, Anderson–Darling, Kolmogorov–Smirnov

$$JB = \frac{n}{6} \left(S^2 + \frac{(K-3)^2}{4} \right) \sim \chi^2(2)$$

- Moment generating function: $M_X(t) = \exp(\mu t + \sigma^2 t^2/2)$
- Multivariate normal and Mahalanobis distance

5.3 PhD Extension

- Berry–Esseen theorem: $\sup_x |F_n(x) - \Phi(x)| \leq \frac{C\rho}{\sigma^3\sqrt{n}}$, where $\rho = \mathbb{E}[|X - \mu|^3]$
- Rate of convergence to normality as a function of the tail index
- Edgeworth expansions for finite-sample corrections
- Normal as $\alpha = 2$ member of the stable family

6 Section 2.4: Stylized Facts of Financial Returns

6.1 Year 3 Core

1. **Fat tails / excess kurtosis:** $\kappa = \mathbb{E}[(X - \mu)^4]/\sigma^4 > 3$; typical daily equity kurtosis: [5, 50]
2. **Negative skewness:** Left tail heavier than right, especially for equity indices
3. **Volatility clustering:** $\text{Corr}(|r_t|, |r_{t+k}|) > 0$ for large k ; “large changes followed by large changes” (Mandelbrot, 1963)
4. **Aggregational Gaussianity:** Kurtosis decreases as holding period increases (daily $\rightarrow$ weekly $\rightarrow$ monthly)

6.2 Master Extension

5. **Leverage effect:** Negative returns increase future volatility more than positive returns of equal magnitude
6. **Absence of return autocorrelation:** $\text{Corr}(r_t, r_{t+k}) \approx 0$ for $k \geq 1$, yet $\text{Corr}(r_t^2, r_{t+k}^2) > 0$
7. **Long memory in volatility:** Slow decay of $\text{Corr}(|r_t|, |r_{t+k}|)$ — hyperbolic, not exponential
8. **Gain/loss asymmetry:** Crashes are sharper than rallies

6.3 PhD Extension

- Cont (2001) complete taxonomy of 11 stylized facts
- Multifractal models (Mandelbrot, Fisher & Calvet, 1997)
- Multiscaling: $\mathbb{E}[|r_t|^q] \propto \Delta t^{\zeta(q)}$ with nonlinear $\zeta(q)$
- Cross-correlation structure and multivariate stylized facts
- Intraday stylized facts: microstructure noise, bid-ask bounce, Epps effect

7 Section 2.5: The Student- t Distribution

7.1 Year 3 Core

Definition 3 (Student- t Distribution). $X \sim t_\nu$ if:

$$f(x; \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}, \quad x \in \mathbb{R}.$$

Key properties:

- Single parameter $\nu > 0$ (degrees of freedom) controls tail heaviness
- $\mathbb{E}[X] = 0$ ($\nu > 1$); $\text{Var}(X) = \frac{\nu}{\nu-2}$ ($\nu > 2$)
- Excess kurtosis = $\frac{6}{\nu-4}$ ($\nu > 4$)
- $\nu \rightarrow \infty$: converges to $N(0, 1)$; $\nu = 1$: Cauchy distribution
- Tails: $f(x) \sim |x|^{-(\nu+1)}$ (power law)
- Financial returns: typically $\hat{\nu} \in [3, 8]$ for daily equity data

7.2 Master Extension

- **Location-scale form:** $X = \mu + \sigma T$ where $T \sim t_\nu$
- MLE estimation of (μ, σ, ν)
- Information criteria: $\text{AIC} = -2\ell + 2k$, $\text{BIC} = -2\ell + k \ln n$
- Student- t as innovation distribution in GARCH models
- QQ-plot diagnostics: systematic deviations indicate misspecification

7.3 PhD Extension

- **Mixture representation:** $X | V \sim N(0, V)$, $V \sim \text{Inv-Gamma}(\nu/2, \nu/2)$
- Bayesian interpretation: Student- t as marginal posterior under conjugate normal-inverse-gamma prior
- Multivariate Student- t : PDF involves Γ functions and determinant
- Comparison with stable distributions: both have power-law tails, but Student- t has finite moments for $\nu > p$
- Tail equivalence: t_ν tail index = ν vs. stable tail index = α

8 Section 2.6: Asymmetric Distributions

8.1 Year 3 Core

- Negative skewness in equity returns: downside risk is larger than upside potential
- **Skewness** measures asymmetry: $S = \mathbb{E}[(X - \mu)^3]/\sigma^3$
- Symmetric distributions (Normal, Student- t) cannot distinguish left from right tail risk
- Intuition: we need distributions with different left and right tail behavior

8.2 Master Extension

Definition 4 (Hansen's Skewed- t Distribution (Hansen, 1994)).

$$f(x; \nu, \lambda) = \begin{cases} bc \left(1 + \frac{1}{\nu-2} \left(\frac{bx+a}{1-\lambda} \right)^2 \right)^{-(\nu+1)/2} & x < -a/b \\ bc \left(1 + \frac{1}{\nu-2} \left(\frac{bx+a}{1+\lambda} \right)^2 \right)^{-(\nu+1)/2} & x \geq -a/b \end{cases}$$

where $\lambda \in (-1, 1)$ is the skewness parameter, $a = 4\lambda c \frac{\nu-2}{\nu-1}$, $b^2 = 1 + 3\lambda^2 - a^2$, $c = \frac{\Gamma((\nu+1)/2)}{\sqrt{\pi(\nu-2)}\Gamma(\nu/2)}$.

- $\lambda = 0$: symmetric Student- t ; $\lambda < 0$: heavier left tail
- **GED**: $f(x; \nu) = \frac{\nu}{2\Gamma(1/\nu)} \exp(-|x|^\nu)$; $\nu = 2$: Normal; $\nu = 1$: Laplace
- Widely used as GARCH innovation distributions

8.3 PhD Extension

- **Generalized Hyperbolic (GH)** distribution: five parameters $(\lambda, \alpha, \beta, \delta, \mu)$
- Special cases: Hyperbolic ($\lambda = 1$), NIG ($\lambda = -1/2$)
- GH as normal mean-variance mixture: $X | W \sim N(\mu + \beta W, W)$, $W \sim \text{GIG}(\lambda, \delta, \gamma)$
- Barndorff-Nielsen (1977): empirical fit to financial data
- Affine closure: GH distributions are closed under affine transformations

9 Section 2.7: Stable (α -Stable) Distributions [MANDATORY]

9.1 Year 3 Core

Definition 5 (Stable Distribution — Intuitive). A distribution is **stable** if a sum of i.i.d. copies has the same shape (up to location and scale). Formally, $X \sim S(\alpha, \beta, \gamma, \delta)$ if for i.i.d. copies X_1, X_2 :

$$aX_1 + bX_2 \stackrel{d}{=} cX + d, \quad c = (a^\alpha + b^\alpha)^{1/\alpha}.$$

- **Four parameters:**

- $\alpha \in (0, 2]$: **stability index** (tail heaviness) — smaller $\alpha \Rightarrow$ heavier tails
- $\beta \in [-1, 1]$: **skewness**
- $\gamma > 0$: **scale**
- $\delta \in \mathbb{R}$: **location**

- **Special cases:** $\alpha = 2$ (Gaussian), $\alpha = 1, \beta = 0$ (Cauchy)

- **Why variance may not exist:** For $\alpha < 2$, $\mathbb{E}[X^2] = \infty$. The sample variance does not converge.

- **Financial motivation:** Mandelbrot (1963) observed that cotton price changes follow stable laws with $\alpha \approx 1.7$

- Typical estimates for financial returns: $\hat{\alpha} \in [1.5, 1.9]$

9.2 Master Extension

Characteristic function ($\alpha \neq 1$):

$$\phi(t) = \mathbb{E}[e^{itX}] = \exp\left\{-\gamma^\alpha |t|^\alpha \left(1 - i\beta \operatorname{sign}(t) \tan \frac{\pi\alpha}{2}\right) + i\delta t\right\}$$

Case $\alpha = 1$:

$$\phi(t) = \exp\left\{-\gamma |t| \left(1 + i\beta \frac{2}{\pi} \operatorname{sign}(t) \ln |t|\right) + i\delta t\right\}$$

- **No closed-form PDF** in general; density via inverse Fourier transform:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi(t) e^{-itx} dt$$

- **Generalized Central Limit Theorem (GCLT):** Stable distributions are the *only* possible non-degenerate limits of normalized sums $\frac{S_n - b_n}{a_n} \xrightarrow{d} S(\alpha, \beta, \gamma, \delta)$

- **Stability property:** Closed under convolution — sums of i.i.d. stable r.v.'s remain stable with the same α

- **Infinite variance** when $\alpha < 2$: $\mathbb{E}[|X|^p] < \infty$ iff $p < \alpha$

- **Power-law tails:** $P(X > x) \sim C_\alpha (1 + \beta) x^{-\alpha}$ as $x \rightarrow \infty$

- **Estimation methods:** McCulloch quantile method, maximum likelihood (Nolan, 1997), characteristic function methods (Kogon & Williams, 1998)

- **Financial interpretation:** α measures tail heaviness; $\alpha \approx 1.7$ for S&P 500 daily returns

9.3 PhD Extension

Definition 6 (Regular Variation). A measurable function $L : (0, \infty) \rightarrow (0, \infty)$ is **slowly varying** at infinity if $\lim_{x \rightarrow \infty} L(tx)/L(x) = 1$ for all $t > 0$. A function f is **regularly varying** with index ρ if $f(x) = x^\rho L(x)$.

Theorem 1 (Domain of Attraction). A distribution F belongs to the **domain of attraction** of a stable law $S(\alpha, \beta, 1, 0)$ if and only if:

1. $\bar{F}(x) = x^{-\alpha} L(x)$ for a slowly varying function L (regularly varying tails)
2. The balance condition: $\lim_{x \rightarrow \infty} \frac{\bar{F}(x)}{\bar{F}(x) + \bar{F}(-x)} = \frac{1+\beta}{2}$
 - **Lévy processes:** A Lévy process $\{L_t\}$ has stationary independent increments; $L_1 \sim S(\alpha, \beta, \gamma, \delta)$ defines a stable Lévy process
 - **Lévy–Itô decomposition:** $L_t = bt + \sigma W_t + \int_{|x| \leq 1} x \tilde{N}(t, dx) + \int_{|x| > 1} x N(t, dx)$
 - **Lévy measure** of a stable process: $\nu(dx) = \begin{cases} c_+ x^{-\alpha-1} dx & x > 0 \\ c_- |x|^{-\alpha-1} dx & x < 0 \end{cases}$
 - **Estimation challenges:**
 - MLE requires numerical evaluation of the density (Nolan’s algorithm)
 - Likelihood surface can be multimodal for small samples
 - Tail index estimators: Hill, Pickands, moment estimator
 - **Tempered stable distributions:** Modify the Lévy measure to ensure finite moments: $\nu(dx) = c |x|^{-\alpha-1} e^{-\lambda|x|} dx$; retain heavy-tail behavior for moderate x but guarantee finite variance
 - **Stable vs. Student- t :** Both have power-law tails, but Student- t has finite moments up to order $\lfloor \nu \rfloor - 1$; stable has no finite moments beyond order α . Student- t is not closed under convolution; stable is. Debate: Mandelbrot–Fama (stable) vs. Praetz–Blattberg–Gonides (Student- t)

10 Section 2.8: Extreme Value Theory (EVT)

10.1 Year 3 Core

- **Motivation:** We need accurate models for extreme events (financial crises, market crashes)
- **Key idea:** Instead of modeling the *entire* distribution, EVT focuses on the **tails** only
- **GPD concept** (descriptive): Exceedances above a high threshold tend to follow a specific distribution shape — the Generalized Pareto Distribution
- **Tail index** ξ (intuitive): a single number that summarizes how “heavy” the tail is
 - $\xi > 0$: heavy tails (power-law decay) — typical for financial data
 - $\xi = 0$: exponential tails (similar to Normal)
 - $\xi < 0$: bounded tails (finite maximum)
- Financial data: $\hat{\xi} > 0$ confirms that extreme events are more frequent than the Normal predicts
- **Note:** The formal theorems and VaR computation from EVT are presented at the Master level

10.2 Master Extension

Theorem 2 (Fisher–Tippett–Gnedenko). If $M_n = \max(X_1, \dots, X_n)$ and $\frac{M_n - b_n}{a_n} \xrightarrow{d} H$, then H is the GEV:

$$H(x; \xi, \mu, \sigma) = \exp \left\{ - \left(1 + \xi \frac{x - \mu}{\sigma} \right)^{-1/\xi} \right\}$$

with $\xi = 0$ (Gumbel), $\xi > 0$ (Fréchet), $\xi < 0$ (Weibull).

Theorem 3 (Pickands–Balkema–de Haan). For high threshold u , exceedances $Y = X - u \mid X > u$ follow the GPD:

$$G(y; \xi, \sigma) = 1 - \left(1 + \xi \frac{y}{\sigma} \right)^{-1/\xi}, \quad y > 0.$$

- **POT method:** Estimate (ξ, σ) from exceedances above threshold u
- **VaR from GPD:**

$$\text{VaR}_\alpha = u + \frac{\sigma}{\xi} \left[\left(\frac{n}{N_u} \cdot \alpha \right)^{-\xi} - 1 \right]$$

where N_u = number of exceedances, n = total observations

- **ES from GPD:** $\text{ES}_\alpha = \frac{\text{VaR}_\alpha}{1-\xi} + \frac{\sigma - \xi u}{1-\xi}$, for $\xi < 1$
- **Threshold selection:** Mean excess plot, parameter stability plot
- **Mean excess function:** $e(u) = \mathbb{E}[X - u \mid X > u]$; linear in u for GPD

10.3 PhD Extension

Definition 7 (Hill Estimator). For order statistics $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)}$, the Hill estimator of the tail index is:

$$\hat{\xi}_{\text{Hill}}^{(k)} = \frac{1}{k} \sum_{i=1}^k \ln X_{(i)} - \ln X_{(k+1)}, \quad k = 1, \dots, n-1.$$

The tail index is $\alpha = 1/\hat{\xi}_{\text{Hill}}$.

- **Hill plot:** $\hat{\xi}_{\text{Hill}}^{(k)}$ vs. k ; stable region indicates appropriate k
- **Bias–variance trade-off:** Small $k \Rightarrow$ high variance; large $k \Rightarrow$ high bias
- **Second-order conditions:** Rate of convergence of the tail to the Pareto form; Hall (1982) second-order parameter ρ governs optimal k
- **Adaptive threshold selection:** Minimizing asymptotic MSE (Danielsson et al., 2001)
- **Pickands estimator:** $\hat{\xi} = \frac{1}{\ln 2} \ln \frac{X_{(k)} - X_{(2k)}}{X_{(2k)} - X_{(4k)}}$ — does not require $\xi > 0$
- **Moment estimator** (Dekkers–Einmahl–de Haan): Works for all $\xi \in \mathbb{R}$
- **Connection to stable distributions:** If $X \sim S(\alpha, \beta, \gamma, \delta)$ with $\alpha < 2$, then $\xi = 1/\alpha$

11 Section 2.9: Distribution Comparison

11.1 Year 3 Core

- Histogram with fitted overlays (Normal, Student- t , Stable)
- QQ-plots: departures from the 45° line
- Log-log tail plot: empirical survival function vs. theoretical
- Visual assessment: which distribution best captures the center, shoulders, and tails?

11.2 Master Extension

- AIC/BIC comparison table across distributions
- Kolmogorov–Smirnov test: $D_n = \sup_x |F_n(x) - F_\theta(x)|$
- Anderson–Darling test: weighted CDF distance, more sensitive to tails
- Likelihood ratio test for nested models (Normal vs. Student- t)

11.3 PhD Extension

- Vuong (1989) test for non-nested model comparison
- Scoring rules: CRPS (Continuous Ranked Probability Score)
- Diebold–Mariano test for comparing predictive densities
- Cross-validation for distributional models
- Probability Integral Transform (PIT) and Rosenblatt test

12 Section 2.10: Risk Management Implications

12.1 Year 3 Core — Introducing VaR from Scratch

Pedagogical note: Students at Year 3 level have **not** encountered VaR or ES before. This section introduces these concepts from first principles.

Step 1: The Risk Question.

A bank holds a portfolio worth \$1,000,000. The fundamental risk question is:

“What is the maximum loss we can expect on a given day, with a specified level of confidence?”

Step 2: Quantile Intuition.

If we know the distribution of daily returns r_t , then the α -quantile q_α is the value such that:

$$P(r_t \leq q_\alpha) = \alpha.$$

For example, if $\alpha = 0.01$, then $q_{0.01}$ is the return level that is exceeded (on the downside) only 1% of the time. In 100 trading days, we expect to lose *more* than $q_{0.01}$ on approximately 1 day.

Step 3: Formal Definition.

Definition 8 (Value-at-Risk — First Introduction). The **Value-at-Risk** at confidence level $(1 - \alpha)$ is the loss threshold that is exceeded with probability α :

$$\text{VaR}_\alpha = -q_\alpha = -F^{-1}(\alpha),$$

where F is the CDF of returns. The minus sign converts the negative quantile into a positive loss amount.

Interpretation: “With probability $(1 - \alpha)$, the daily loss will **not exceed** VaR_α .”

Step 4: VaR under the Normal Assumption.

If $r_t \sim N(\mu, \sigma^2)$, then:

$$\text{VaR}_\alpha = -(\mu + \sigma \cdot z_\alpha), \quad \text{where } z_\alpha = \Phi^{-1}(\alpha).$$

Example: $\mu = 0$, $\sigma = 0.015$ (daily), portfolio = \$1M, $\alpha = 0.01$:

$$\text{VaR}_{0.01} = -0.015 \times (-2.326) \times 1,000,000 = \$34,890.$$

Interpretation: “On 99% of trading days, the loss will not exceed \$34,890.”

Step 5: Why the Distribution Matters.

Under a Student- t with $\nu = 5$, the 1% quantile is more extreme than -2.326 . So VaR is **higher**:

Model	99% VaR (\$1M)	99.9% VaR (\$1M)	Ratio to Normal
Normal	\$27,000	\$35,900	1.0×
Student- t ($\nu = 5$)	\$33,700	\$58,400	1.6×
Stable ($\alpha = 1.7$)	\$41,000	\$90,200	2.5×

- At the 99% level, the difference is moderate ($\sim 25\%$)
- At the 99.9% level, the difference is **dramatic**: the Normal underestimates risk by a factor of 1.6–2.5×
- **Conclusion:** The choice of distribution directly affects how much capital a bank must hold

Step 6: Graphical Intuition.

A slide should show the left tail of the PDF with a shaded area $= \alpha$, and the VaR marked as the boundary point. Two overlaid distributions (Normal vs. Student- t) make the difference visually obvious.

Step 7: Limitations of VaR (preview).

- VaR tells us the *threshold* of the worst α -fraction of days, but not *how bad* things get beyond that threshold
- VaR does not distinguish between “losing \$35k” and “losing \$200k” — both are simply “VaR exceedances”
- A better measure (Expected Shortfall) will be introduced at the Master level

12.2 Master Extension — Expected Shortfall and Backtesting

Definition 9 (Expected Shortfall — First Introduction at Master Level). The **Expected Shortfall** (also called Conditional VaR) at level α is the average loss in the worst α -fraction of scenarios:

$$\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha F^{-1}(p) dp = \mathbb{E}[-X \mid X \leq -\text{VaR}_\alpha].$$

Interpretation: “Given that we are in the worst $\alpha\%$ of days, what is the *average* loss?”

- ES is a **coherent** risk measure (subadditive): $\text{ES}(X + Y) \leq \text{ES}(X) + \text{ES}(Y)$
- VaR is **not** subadditive — diversification can increase VaR
- Basel III/IV: shift from 99% VaR to 97.5% ES
- Kupiec test: $LR_{uc} = -2 \ln \frac{\alpha^x (1-\alpha)^{n-x}}{\hat{\alpha}^x (1-\hat{\alpha})^{n-x}} \sim \chi^2(1)$
- Christoffersen test: independence of exceedances
- Basel traffic light system: Green (0–4), Yellow (5–9), Red (10+) exceedances per 250 days

12.3 PhD Extension

- **Spectral risk measures:** $\rho_\phi(X) = -\int_0^1 \phi(p) F^{-1}(p) dp$ with risk spectrum ϕ
- **Elicitability:** VaR is elicitable (can be backtested via scoring functions); ES is not elicitable alone but jointly elicitable with VaR (Fissler & Ziegel, 2016)
- **Capital allocation:** Euler allocation under ES — distributional assumptions directly affect allocated capital per desk
- **Stable distributions and extreme risk:** ES_α under stable distributions diverges as $\alpha \rightarrow 0$ faster than under Student- t ; implications for capital adequacy
- **Model risk quantification:** Range of risk measures across plausible distributional models

13 Section 2.11: Modern Extensions

13.1 Year 3 Core

- GARCH intuition: variance changes over time; yesterday’s large shock $\Rightarrow$ today’s high variance
- $\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ (conceptual formula)
- After GARCH filtering, standardized residuals are closer to Gaussian but still show excess kurtosis

13.2 Master Extension

- GARCH(1,1) with Student- t innovations: $r_t = \mu + \sigma_t z_t$, $z_t \sim t_\nu$
- Gaussian mixtures: $f(x) = \sum_{k=1}^K \pi_k \phi(x; \mu_k, \sigma_k^2)$; calm vs. crisis regimes
- EM algorithm for mixture estimation
- Hamilton (1989) regime-switching: Markov chain latent state s_t
- Asymmetric GARCH: GJR-GARCH, EGARCH for leverage effect

13.3 PhD Extension

- Stochastic volatility: $d \ln \sigma_t^2 = \kappa(\theta - \ln \sigma_t^2) dt + \sigma_v dW_t^v$ with correlated Brownian motions
- Jump-diffusion models: Merton (1976), Kou (2002)
- Lévy-driven volatility models: BNS model (Barndorff-Nielsen & Shephard, 2001)
- Realized volatility and high-frequency estimation
- GARCH with stable innovations: Mittnik & Rachev (1993)
- Fractionally integrated GARCH (FIGARCH) for long memory

Part III

KEY FORMULAS

14 Complete Formula Reference

14.1 Returns and Moments

1. **Log-return:** $r_t = \ln(P_t/P_{t-1})$
2. **Sample mean:** $\bar{r} = \frac{1}{n} \sum_{t=1}^n r_t$
3. **Sample variance:** $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2$
4. **Skewness:** $S = \frac{1}{n} \sum_{t=1}^n \left(\frac{r_t - \bar{r}}{\hat{\sigma}} \right)^3$
5. **Kurtosis:** $K = \frac{1}{n} \sum_{t=1}^n \left(\frac{r_t - \bar{r}}{\hat{\sigma}} \right)^4$
6. **Excess kurtosis:** $K_e = K - 3$

14.2 Distribution PDFs

7. **Normal PDF:** $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
8. **Student- t PDF:** $f(x; \nu) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}$
9. **Student- t excess kurtosis:** $\kappa_e = \frac{6}{\nu-4}, \quad \nu > 4$
10. **Hansen's Skewed- t PDF:** see Section 2.6
11. **GED PDF:** $f(x; \nu) = \frac{\nu}{2\Gamma(1/\nu)} \exp(-|x|^\nu)$

14.3 Stable Distributions

12. **Stable characteristic function** ($\alpha \neq 1$):

$$\phi(t) = \exp\left\{-\gamma^\alpha |t|^\alpha \left(1 - i\beta \operatorname{sign}(t) \tan \frac{\pi\alpha}{2}\right) + i\delta t\right\}$$

13. **Moment existence:** $\mathbb{E}[|X|^p] < \infty$ iff $p < \alpha$

14. **Tail behavior:** $P(X > x) \sim C_\alpha (1 + \beta) x^{-\alpha}, \quad C_\alpha = \frac{1-\alpha}{\Gamma(2-\alpha) \cos(\pi\alpha/2)}$

14.4 Extreme Value Theory

15. **GEV CDF**: $H(x; \xi, \mu, \sigma) = \exp\left\{-\left(1 + \xi \frac{x-\mu}{\sigma}\right)^{-1/\xi}\right\}$
16. **GPD CDF**: $G(y; \xi, \sigma) = 1 - \left(1 + \xi \frac{y}{\sigma}\right)^{-1/\xi}, \quad y > 0$
17. **Hill estimator**: $\hat{\xi}_{\text{Hill}}^{(k)} = \frac{1}{k} \sum_{i=1}^k \ln X_{(i)} - \ln X_{(k+1)}$
18. **Mean excess function**: $e(u) = \mathbb{E}[X - u \mid X > u] = \frac{\sigma + \xi u}{1 - \xi} \text{ (GPD)}$

14.5 Risk Measures

19. **VaR [Y3]** (introduced in Section 2.10): $\text{VaR}_\alpha = -F^{-1}(\alpha)$
20. **VaR under normality [Y3]**: $\text{VaR}_\alpha = -(\mu + \sigma \Phi^{-1}(\alpha))$
21. **ES under normality [MSc]**: $\text{ES}_\alpha = -\mu + \sigma \frac{\phi(\Phi^{-1}(\alpha))}{\alpha}$
22. **VaR from GPD [MSc]**: $\text{VaR}_\alpha = u + \frac{\sigma}{\xi} \left[\left(\frac{n\alpha}{N_u} \right)^{-\xi} - 1 \right]$
23. **ES from GPD [MSc]**: $\text{ES}_\alpha = \frac{\text{VaR}_\alpha}{1 - \xi} + \frac{\sigma - \xi u}{1 - \xi}$
24. **Kupiec LR test [MSc]**: $LR_{uc} = -2 \ln \frac{\alpha^x (1 - \alpha)^{n-x}}{\hat{\alpha}^x (1 - \hat{\alpha})^{n-x}} \sim \chi^2(1)$

14.6 Model Selection

25. **AIC**: $\text{AIC} = -2\ell + 2k$
26. **BIC**: $\text{BIC} = -2\ell + k \ln n$
27. **Jarque–Bera**: $JB = \frac{n}{6} \left(S^2 + \frac{(K-3)^2}{4} \right) \sim \chi^2(2)$

All symbols are defined where first introduced:

- P_t : asset price; r_t : log-return; n : sample size
- μ : location/mean; σ : scale/std dev; ν : degrees of freedom
- α : stability index (stable) or tail probability (VaR)
- β : skewness (stable); γ : scale (stable); δ : location (stable)
- ξ : shape parameter (GEV/GPD); ℓ : log-likelihood; k : number of parameters
- Φ : standard normal CDF; ϕ : standard normal PDF; Γ : gamma function

Part IV

EMPIRICAL ILLUSTRATIONS

15 Required Graphs — Mapping to Existing Charts

#	Required Graph	Existing File	Layer
G1	Price series + log-return plot	ch2_normal_fit.png	Y3
G2	Histogram + Normal fit overlay	ch2_normal_fit.png	Y3
G3	Histogram + Student- t fit	ch2_student_t_fit.png	Y3
G4	Student- t PDFs (varying ν)	ch2_student_t_pdfs.png	Y3
G5	Histogram + Stable fit comparison	ch2_sp500_stable_fit.png	Y3/MSc
G6	Stable PDFs (varying α)	ch2_stable_pdfs.png	MSc
G7	QQ-plot vs. Normal	ch2_normality_tests.png	Y3
G8	QQ-plot vs. Student- t	ch2_dist_comparison_qq.png	Y3
G9	Log-log tail plot	ch2_dist_comparison_tails.png	MSc
G10	Distribution overlay comparison	ch2_dist_comparison_overlay.png	Y3
G11	CLT illustration	ch2_clt_illustration.png	Y3
G12	GEV fit to block maxima	ch2_evt_gev_fit.png	MSc
G13	GPD / POT analysis	ch2_evt_gpd_pot.png	MSc
G14	EVT-based VaR and ES	ch2_evt_var_es.png	MSc
G15	VaR comparison bar chart	ch2_var_comparison.png	Y3
G16	VaR backtest	ch2_var_backtest.png	MSc
G17	Risk comparison table	ch2_risk_comparison_table.png	Y3/MSc
G18	GARCH- t fit	ch2_garch_t_fit.png	MSc
G19	Cross-asset stability analysis	ch2_cross_asset_alpha.png	MSc/PhD
G20	Tail survival probability plot	ch2_tail_survival.png	PhD

15.1 Additional Graphs Needed

#	Graph	Layer
G21	Hill plot ($\hat{\xi}$ vs. k)	PhD
G22	Mean excess plot with threshold line	MSc/PhD
G23	Stable vs. Student- t tail comparison (log-log)	PhD
G24	VaR graphical intuition: PDF with shaded tail and VaR threshold (TikZ)	Y3
G24b	VaR exceedance count by model (bar chart)	Y3/MSc
G25	Aggregational Gaussianity: kurtosis vs. horizon	Y3/MSc
G26	QQ-plot vs. Stable distribution	MSc
G27	Regime classification plot	MSc
G28	Gaussian mixture fit	MSc

Part V

QUANTLET PLAN

16 Quantlet Architecture

16.1 Q1: Stylized Facts Analysis

Name	SFM_ch2_stylized_facts
Layer	[Y3] Year 3 Core
Objective	Demonstrate the key stylized facts of financial returns: fat tails, skewness, volatility clustering, aggregational Gaussianity
Data	S&P 500 daily prices from Yahoo Finance (<code>yfinance</code>), 2000–2025
Steps	1. Download prices, compute log-returns 2. Plot price series and return series side by side 3. Compute sample moments: mean, std, skewness, kurtosis 4. Histogram with Normal overlay 5. QQ-plot vs. Normal 6. ACF of returns vs. ACF of r_t and r_t^2 (volatility clustering) 7. Kurtosis at different horizons (daily, weekly, monthly)
Graphs	G1, G2, G7, G11, G25
Interpretation	Report: normality rejected (Jarque–Bera), kurtosis $\gg 3$, volatility clustering evident, convergence to normality at lower frequencies

16.2 Q2: Normal vs. Student- t Fit

Name	SFM_ch2_normal_vs_student
Layer	[Y3] Year 3 Core
Objective	Fit Normal and Student- t distributions via MLE; compare goodness of fit
Data	S&P 500 daily log-returns
Steps	1. MLE fit of $N(\mu, \sigma^2)$ 2. MLE fit of location-scale $t_\nu(\mu, \sigma)$ 3. Report parameters: $(\hat{\mu}, \hat{\sigma})$ and $(\hat{\mu}, \hat{\sigma}, \hat{\nu})$ 4. Compute AIC, BIC for both models 5. Histogram with both fitted PDFs overlaid 6. QQ-plots: Normal and Student-t
Graphs	G2, G3, G4, G7, G8, G10
Interpretation	Student- t achieves substantially better AIC/BIC despite extra parameter; QQ-plot improvement in tails; typical $\hat{\nu} \in [3, 6]$

16.3 Q3: VaR Comparison

Name	SFM_ch2_var_comparison
Layer	[Y3] Year 3 / [MSc] Master
Objective	Compute and compare VaR and ES under Normal, Student- t , and empirical quantile approaches
Data	S&P 500 daily log-returns
Steps	1. Fit Normal and Student-t distributions 2. Compute VaR at 95%, 99%, 99.9% under each model 3. Compute empirical VaR (historical simulation) 4. Compute ES under each model 5. Bar chart comparing all VaR estimates 6. Table with \$1M portfolio loss estimates 7. [MSc] Rolling window VaR backtest (250-day window) 8. [MSc] Kupiec test for each model
Graphs	G15, G16, G17, G24
Interpretation	Normal severely underestimates tail risk at extreme quantiles; Student- t and empirical approaches are more conservative; backtesting confirms Normal has too many violations

16.4 Q4: Tail Estimation (Hill Estimator)

Name	SFM_ch2_tail_estimation
Layer	[MSc] Master / [PhD] PhD
Objective	Estimate the tail index using the Hill estimator and compare with parametric tail indices
Data	S&P 500 daily absolute log-returns
Steps	1. Sort absolute returns in descending order 2. Compute Hill estimator $\hat{\xi}^{(k)}$ for $k = 10, 11, \dots, 500$ 3. Plot Hill plot: $\hat{\alpha} = 1/\hat{\xi}$ vs. k 4. Identify stable region for tail index estimate 5. Compare with $\hat{\alpha}$ from stable fit and $\hat{\nu}$ from Student-t fit 6. [PhD] Pickands and moment estimators for comparison 7. [PhD] Bias-corrected Hill estimator
Graphs	G20, G21, G23
Interpretation	Hill estimate typically gives $\hat{\alpha} \in [3, 5]$ (corresponding to Student- t with $\nu \approx 3-5$); stable fit gives $\hat{\alpha} \in [1.5, 1.9]$; discuss the difference between tail index and stability index

16.5 Q5: Stable Distribution Simulation and Fitting

Name	SFM_ch2_stable_distributions
Layer	[Y3] Year 3 / [MSc] Master
Objective	Simulate from stable distributions; fit stable model to S&P 500 returns
Data	Simulated data + S&P 500 daily returns
Steps	1. Simulate $S(\alpha, 0, 1, 0)$ for $\alpha = 1.2, 1.5, 1.7, 2.0$ 2. Plot PDFs on linear and log scales 3. Demonstrate stability property: $X_1 + X_2 \stackrel{d}{=} cX + d$ 4. [MSc] Fit stable distribution to S&P 500 via MLE (<code>scipy.stats.levy_stable</code>) 5. [MSc] Report $(\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \hat{\delta})$ 6. [MSc] Compare stable fit vs. Normal on log-scale plot 7. [MSc] Cross-asset analysis: estimate $\hat{\alpha}$ for multiple assets
Graphs	G5, G6, G19, G26
Interpretation	Stable distributions capture heavy tails better than Normal; estimated $\hat{\alpha} < 2$ confirms infinite-variance regime; discuss implications for portfolio theory

16.6 Q6: EVT-Based VaR

Name	SFM_ch2_evt_var
Layer	[MSc] Master / [PhD] PhD
Objective	Apply EVT (POT method) to estimate tail risk measures
Data	S&P 500 daily losses (negated returns)
Steps	1. Compute losses $L_t = -r_t$ 2. Mean excess plot: $\hat{e}(u)$ vs. u 3. Select threshold u from linearity region 4. Fit GPD to exceedances via MLE 5. Report $(\hat{\xi}, \hat{\sigma})$; confirm $\hat{\xi} > 0$ 6. Compute EVT-based VaR and ES at 99%, 99.9% 7. Compare with Normal, Student-t, and historical simulation 8. [PhD] GEV fit to monthly block maxima 9. [PhD] Hill plot and tail index comparison 10. [PhD] Parameter stability plot across thresholds
Graphs	G12, G13, G14, G22
Interpretation	EVT provides theoretically justified extrapolation beyond observed data; GPD-based VaR is more conservative than Normal but less extreme than Stable; discuss threshold sensitivity

Part VI

EXERCISES

17 Exercises — Three Difficulty Levels

17.1 Year 3 Exercises (Undergraduate)

E1. Log>Returns Computation. Download 5 years of daily closing prices for Apple (AAPL) from Yahoo Finance.

- (a) Compute the daily log-returns.
- (b) Plot the price series and the return series.
- (c) Compute the sample mean, standard deviation, skewness, and kurtosis.
- (d) Is the excess kurtosis significantly different from zero? What does this imply?

E2. Normal vs. Student- t Fit. Using the AAPL returns from Exercise 1:

- (a) Fit a Normal distribution by MLE and report $(\hat{\mu}, \hat{\sigma})$.
- (b) Fit a Student- t distribution by MLE and report $(\hat{\mu}, \hat{\sigma}, \hat{\nu})$.
- (c) Produce a histogram with both fitted PDFs overlaid.
- (d) Produce QQ-plots against both distributions. Which fits better in the tails?
- (e) Compute AIC and BIC for both models. Which model is preferred?

E3. Introduction to VaR. This exercise introduces Value-at-Risk (VaR) step by step.

- (a) In your own words, explain what the statement “the 99% VaR is \$25,000” means for a bank.
- (b) Using the AAPL returns from Exercise 1, compute the sample mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$.
- (c) Under the Normal assumption, the 1% quantile is $q_{0.01} = \hat{\mu} + \hat{\sigma} \times (-2.326)$. Compute this value.
- (d) The VaR is $\text{VaR}_{0.01} = -q_{0.01}$. Compute it. For a \$100,000 portfolio, what is the maximum daily loss at 99% confidence?
- (e) Now compute the 1% *empirical* quantile directly from the data (i.e., sort all returns and find the value below which 1% of observations fall). Compare with the Normal-based VaR. Which is larger? What does the difference tell you about the Normal assumption?

E4. Stylized Facts. Using S&P 500 daily returns (2000–2024):

- (a) Compute the kurtosis at daily, weekly, and monthly frequencies.
- (b) Does the kurtosis decrease with the holding period? This is called *aggregational Gaussianity* — explain why it occurs.
- (c) Plot the ACF of returns and the ACF of absolute returns. What is the key difference? Relate to volatility clustering.

E5. Stable Distributions: Conceptual.

- (a) What does it mean for a distribution to be “stable”?
- (b) Name three special cases of the stable distribution family.
- (c) If $\alpha = 1.7$, does the variance exist? Does the mean exist? Explain.
- (d) Why did Mandelbrot (1963) advocate stable distributions for financial returns?

17.2 Master Exercises

M1. Comprehensive Distribution Fitting. Download 20 years of daily returns for a major equity index (e.g., S&P 500, DAX, or Nikkei 225).

- (a) Fit the following distributions by MLE: Normal, Student- t , Skewed- t (Hansen), Stable.
- (b) Report all fitted parameters in a table.

- (c) Compute AIC, BIC, and log-likelihood for each model.
- (d) Produce a panel of QQ-plots (one per distribution).
- (e) On a log-scale histogram, overlay all fitted PDFs. Which model captures the tails best?
- (f) Discuss: Is the Stable distribution's infinite variance a practical problem?

M2. VaR Backtesting. Using a rolling window of 250 trading days:

- (a) Compute the 1-day ahead 99% VaR under Normal, Student- t , and historical simulation.
- (b) Count the number of VaR exceedances for each model over the out-of-sample period.
- (c) Apply the Kupiec test to each model. Which models pass at the 5% significance level?
- (d) Apply the Christoffersen test. Are exceedances independent?
- (e) Classify each model under the Basel traffic light system.

M3. EVT Application.

- (a) For S&P 500 daily losses, plot the mean excess function.
- (b) Select an appropriate threshold and justify your choice.
- (c) Fit the GPD to exceedances and report $(\hat{\xi}, \hat{\sigma})$.
- (d) Compute the EVT-based 99% and 99.9% VaR and ES.
- (e) Compare with Normal-based and Student- t -based estimates.
- (f) Fit the GEV distribution to monthly block maxima and compare $\hat{\xi}$.

M4. GARCH- t Filtering.

- (a) Fit a GARCH(1,1) model with Normal innovations to S&P 500 returns.
- (b) Fit a GARCH(1,1) model with Student- t innovations.
- (c) Compare the models using AIC/BIC and the log-likelihood.
- (d) Extract standardized residuals from the GARCH- t model. Are they closer to Gaussian than the raw returns?
- (e) Compute the kurtosis of the standardized residuals. How much of the original excess kurtosis is “explained” by GARCH?

M5. Cross-Asset Analysis. Estimate the stability index $\hat{\alpha}$ for five different assets (e.g., S&P 500, EUR/USD, Gold, Bitcoin, 10-Year Treasury).

- (a) Report the estimated parameters in a table.
- (b) Which asset class has the heaviest tails?
- (c) Is $\hat{\alpha}$ significantly different from 2 for any of the assets?
- (d) Discuss implications for portfolio diversification.

17.3 PhD Exercises

P1. Tail Index Estimation. For S&P 500 daily absolute returns:

- (a) Compute the Hill estimator $\hat{\xi}^{(k)}$ for $k = 10, \dots, 500$.
- (b) Plot the Hill plot and identify a stable region.
- (c) Compute the Pickands and moment estimators. Do they agree?

- (d) Derive the asymptotic variance of the Hill estimator: $\text{Var}(\hat{\xi}^{(k)}) \sim \xi^2/k$ under second-order conditions.
- (e) Apply the Danielsson et al. (2001) adaptive method for optimal k selection.
- (f) Compare $\hat{\alpha}_{\text{Hill}} = 1/\hat{\xi}$ with the $\hat{\alpha}$ from the stable fit. Discuss.

P2. Domain of Attraction.

- (a) Show that the Student- t_ν distribution is in the domain of attraction of a stable law with index $\alpha = \nu$.
- (b) Prove that the normal distribution is in the domain of attraction of $S(2, 0, \gamma, 0)$.
- (c) For the Cauchy distribution, verify directly that $\bar{F}(x) \sim (\pi x)^{-1}$ is regularly varying with index -1 .
- (d) Discuss: If returns follow a Student- t_5 , what is the appropriate normalizing sequence (a_n, b_n) for the maximum?

P3. Lévy Process Simulation.

- (a) Simulate a symmetric stable Lévy process L_t with $\alpha = 1.7$ at times $t = 1/252, 2/252, \dots, 1$.
- (b) Verify the self-similarity property: $L_{ct} \stackrel{d}{=} c^{1/\alpha} L_t$.
- (c) Simulate a tempered stable process with exponential tempering $e^{-\lambda|x|}$ for $\lambda = 0.1, 1, 10$.
- (d) Compare the tail behavior of the tempered stable with the pure stable.
- (e) Compute sample moments at different time scales; verify that the tempered stable has finite variance but the pure stable does not.

P4. Scoring Rules and Model Comparison.

- (a) Implement the Continuous Ranked Probability Score (CRPS) for each fitted distribution.
- (b) Compute the log-score (predictive log-likelihood) out of sample.
- (c) Apply the Diebold–Mariano test to compare Normal vs. Student- t predictive densities.
- (d) Apply the Vuong test to compare Student- t vs. Stable (non-nested models).
- (e) Implement the Probability Integral Transform (PIT) and test for uniformity.

P5. Spectral Risk Measures.

- (a) Define the spectral risk measure $\rho_\phi(X) = -\int_0^1 \phi(p) F^{-1}(p) dp$ for a risk spectrum ϕ .
- (b) For the exponential spectrum $\phi(p) = \frac{ke^{-k(1-p)}}{1-e^{-k}}$, compute ρ_ϕ under Normal and Student- t assumptions.
- (c) Show that VaR and ES are special cases of spectral risk measures.
- (d) Discuss elicibility: why is ES not elicitable alone? What does joint elicibility (Fissler & Ziegel, 2016) mean for backtesting?

Part VII

EXAM QUESTIONS

18 Year 3 Exam Questions

- Y3-Q1. (25 points)** Consider daily log-returns of a stock with sample statistics: $\bar{r} = 0.0003$, $\hat{\sigma} = 0.015$, skewness = -0.4 , kurtosis = 7.2 .

- (a) (5 pts) Define the log-return and explain why it is preferred over simple returns in statistical analysis.
- (b) (5 pts) Compute the 99% VaR under the Normal assumption for a \$1,000,000 portfolio. Use $z_{0.01} = -2.326$.
- (c) (5 pts) The excess kurtosis is $7.2 - 3 = 4.2$. What does this imply about the tails of the return distribution?
- (d) (5 pts) A Student- t distribution is fitted with $\hat{\nu} = 5$. Verify that the theoretical excess kurtosis is $6/(5 - 4) = 6$. Why might the sample excess kurtosis (4.2) differ from this?
- (e) (5 pts) Which distributional model (Normal or Student- t) would you recommend for computing risk measures? Justify.

Y3-Q2. (25 points)

- (a) (8 pts) State the Central Limit Theorem precisely. What conditions are required?
- (b) (5 pts) Explain “aggregational Gaussianity” in the context of financial returns.
- (c) (7 pts) A Student- t distribution with $\nu = 3$ has infinite kurtosis. Does the CLT apply to sums of i.i.d. t_3 random variables? Explain carefully. (Hint: the CLT requires finite variance.)
- (d) (5 pts) What is the Generalized CLT, and how does it relate to stable distributions?

Y3-Q3. (25 points)

- (a) (7 pts) Define Value-at-Risk at the α level. What does the statement “the 99% VaR is \$50,000” mean in plain language? Draw a diagram showing a PDF with the VaR threshold and the shaded tail area.
- (b) (8 pts) For a portfolio with returns $\sim N(0.001, 0.02^2)$, compute the 95% and 99% VaR for a \$1M investment. Use $z_{0.05} = -1.645$, $z_{0.01} = -2.326$. Show all steps.
- (c) (5 pts) Now suppose the returns follow a Student- t with $\nu = 5$, same location and scale. Without computing exact values, explain *qualitatively* why the VaR would be higher than under the Normal assumption. (Hint: compare the tails of the two distributions.)
- (d) (5 pts) State one important limitation of VaR as a risk measure. (Hint: VaR tells you the threshold of the worst $\alpha\%$ of days, but what does it *not* tell you?)

19 Master Exam Questions

MSc-Q1. (30 points) You are fitting distributions to 5,000 daily returns of the DAX index.

- (a) (5 pts) Write the log-likelihood function for the location-scale Student- t model with parameters (μ, σ, ν) .
- (b) (5 pts) The Normal fit gives $\ell_N = 14,200$ (with $k = 2$). The Student- t fit gives $\ell_t = 14,450$ (with $k = 3$). Compute the AIC and BIC for both models ($n = 5,000$). Which model is preferred?
- (c) (8 pts) Write the characteristic function of the stable distribution $S(\alpha, \beta, \gamma, \delta)$ for $\alpha \neq 1$. What is the stability property? Prove that $S + S' \stackrel{d}{=} cS + d$ for i.i.d. stable copies using the characteristic function.
- (d) (7 pts) Explain the Peaks Over Threshold (POT) method. Write the GPD CDF and derive the VaR formula from the GPD.

- (e) (5 pts) You estimate $\hat{\xi} = 0.25$ from the GPD fit. What does this imply about the tail behavior? Is the variance of the exceedances finite?

MSc-Q2. (30 points) VaR backtesting and regulatory framework.

- (a) (5 pts) Define the Kupiec (unconditional coverage) test. State the null hypothesis, the test statistic, and its asymptotic distribution.
- (b) (8 pts) Over 250 trading days, a bank's 99% VaR model produces 8 exceedances. Compute the Kupiec test statistic. Is the model rejected at the 5% level? ($\chi_{0.05}^2(1) = 3.84$.)
- (c) (7 pts) Classify this result under the Basel traffic light system. What are the regulatory consequences?
- (d) (5 pts) What additional information does the Christoffersen (independence) test provide beyond the Kupiec test?
- (e) (5 pts) Explain why Expected Shortfall was adopted over VaR in Basel III/IV. What is "coherence" and why does VaR fail it?

MSc-Q3. (30 points) Distribution comparison and risk management.

- (a) (8 pts) You estimate the following for S&P 500 returns: $\hat{\alpha}_{\text{stable}} = 1.72$, $\hat{\nu}_{\text{Student-}t} = 4.8$, $\hat{\xi}_{\text{GPD}} = 0.22$. Explain what each parameter measures. Are the three estimates of "tail heaviness" consistent? (Hint: for the GPD, $\xi \approx 1/\alpha_{\text{tail}}$.)
- (b) (7 pts) Compute the 99.9% VaR under the Normal assumption with $\mu = 0$, $\sigma = 0.012$, using $z_{0.001} = -3.09$. Then compute it under the Student- t with $\nu = 5$ using $t_{5,0.001} = -4.77$. Compute the ratio.
- (c) (8 pts) Explain the difference between the tail index estimated by the Hill estimator and the stability index α of a stable distribution. Can both be estimated from the same data? What do they measure differently?
- (d) (7 pts) A GARCH(1,1) model with Normal innovations produces standardized residuals with excess kurtosis 1.5 (down from 18 in the raw returns). Interpret this result. Does it justify using Student- t innovations?

20 PhD Exam Questions

PhD-Q1. (40 points) Stable distributions and domains of attraction.

- (a) (8 pts) State the Generalized Central Limit Theorem. Under what conditions does the limit of normalized sums converge to a stable distribution with $\alpha < 2$?
- (b) (10 pts) Prove that if F has a regularly varying tail with index $-\alpha$ (i.e., $\bar{F}(x) = x^{-\alpha}L(x)$), then F is in the domain of attraction of a stable law $S(\alpha, \beta, 1, 0)$.
- (c) (10 pts) Consider the Student- t distribution with ν degrees of freedom. Show that $P(X > x) \sim c_{\nu} x^{-\nu}$ as $x \rightarrow \infty$ and identify the constant c_{ν} . Hence determine the domain of attraction of t_{ν} .
- (d) (7 pts) Explain the Mandelbrot–Fama vs. Praetz–Blattberg–Gonides debate. What are the practical implications of choosing stable vs. Student- t for risk management?
- (e) (5 pts) Define tempered stable distributions. How do they resolve the infinite-variance issue while retaining heavy-tail behavior?

PhD-Q2. (40 points) Extreme value theory and tail estimation.

- (a) (8 pts) State the Fisher–Tippett–Gnedenko theorem. Derive the three types (Gumbel, Fréchet, Weibull) as special cases of the GEV.
- (b) (10 pts) Prove the Pickands–Balkema–de Haan theorem: for a distribution F in the domain of attraction of H_ξ , the excess distribution $F_u(y) = P(X - u \leq y \mid X > u)$ converges to $G_{\xi, \sigma(u)}(y)$ as $u \rightarrow x^*$.
- (c) (8 pts) Derive the asymptotic distribution of the Hill estimator: $\sqrt{k}(\hat{\xi}_H^{(k)} - \xi) \xrightarrow{d} N(0, \xi^2)$ under appropriate conditions. State the required second-order regular variation condition.
- (d) (7 pts) Explain the bias–variance trade-off in threshold selection for the Hill estimator. How does the Danielsson–de Haan–Peng–Vries (2001) method address this?
- (e) (7 pts) Discuss the connection between the GPD shape parameter ξ and the stable tail index α : if $X \sim S(\alpha, \beta, \gamma, \delta)$, what is ξ in the GPD approximation to the upper tail?

PhD-Q3. (40 points) Advanced risk measurement.

- (a) (8 pts) Define a **coherent risk measure** (Artzner et al., 1999). State the four axioms. Show that VaR violates subadditivity with a counterexample.
- (b) (8 pts) Define spectral risk measures. Show that ES is a spectral risk measure with constant spectrum $\phi(p) = 1/\alpha$ for $p \leq \alpha$.
- (c) (8 pts) Discuss elicibility in the context of risk measure backtesting. Why is ES not elicitable alone? State the Fissler–Ziegel (2016) joint elicibility result for (VaR, ES).
- (d) (8 pts) For a portfolio with returns following a stable distribution $S(1.7, -0.2, 0.01, 0)$, discuss the computation of VaR and ES. What special difficulties arise from the lack of closed-form PDF? How can they be addressed numerically?
- (e) (8 pts) Compare the regulatory approaches of Basel II (99% VaR) and Basel III/IV (97.5% ES). Under which distributional assumptions do these two approaches yield approximately the same capital requirement? Under which do they differ most?

Part VIII

RISK MANAGEMENT CONNECTION

21 Underestimation of Tail Risk Under Normality

Event	σ -Multiple	P under Normal	Empirical Frequency
3σ event	3.0	0.13%	$\sim 1\%$ (daily)
5σ event	5.0	3×10^{-7}	$\sim 0.1\%$ (daily)
10σ event	10.0	$\sim 10^{-23}$	Observed (Oct 1987)
25σ event	25.0	$\sim 10^{-135}$	Observed (Oct 1987)

22 Impact on VaR and ES

- At the **95% level**: All models give similar VaR estimates; distribution choice matters little.

- At the **99% level**: Student- t gives 20–30% higher VaR than Normal.
- At the **99.9% level**: Stable gives $2.5\times$ higher VaR; EVT gives $2.1\times$ higher VaR.
- **ES amplifies the differences**: ES captures the *severity* of tail losses, not just their threshold.
- Under stable distributions with $\alpha < 2$, ES can be dramatically larger because the conditional tail expectation is infinite in the limit.

23 Stable Distributions and Extreme Risk

- If returns truly follow a stable distribution with $\alpha < 2$:
 - The variance does not exist — traditional Markowitz optimization is meaningless.
 - The CLT does not apply — aggregation does not lead to normality.
 - Instead, the GCLT applies: sums converge to a stable law.
 - Risk measures based on variance (Sharpe ratio, tracking error) are unreliable.
- **Practical resolution**: Tempered stable distributions maintain heavy-tail behavior for moderate extremes while ensuring finite variance.

24 Implications for Capital Allocation

- Under-capitalization under Normal assumptions can lead to systemic fragility.
- Basel III/IV shift to ES partially addresses this by focusing on tail severity.
- The choice between Student- t , Stable, and EVT-based risk measures directly affects:
 - Market risk capital requirements
 - Desk-level risk limits
 - Stress testing scenarios
 - Credit risk modeling (default correlation in the tails)
- **Recommendation**: Use Student- t or GARCH- t for day-to-day risk management; use EVT for stress testing and extreme quantile estimation; consider stable distributions for theoretical understanding and research.

Part IX

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25 Core References

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